

Accessibility Palette Sheets



Companion to the Branding Style Sheet: the institute's palette re-issued for each condition it serves. Each sheet is labeled by the condition it responds to, and every value is computed against WCAG 2.1, not estimated.

PROPOSED FOR REVIEW · Drew Stone holds color authority; adoption follows his review · Owner: Branding & Outreach

THE BASE PALETTE THESE SHEETS ADAPT

The settled assignments. Divisions first, Bridge Groups second. Every sheet that follows starts from these values and changes only what the named condition requires.

HOME	COLOR	HEX
Research	 Inquiry Teal	#1A6073
Technology, AI & Information Systems	 Interface Blue	#075BA8
Experience Architecture & World Building	 Embodied Coral	#D15B32
Lumina Lingua	 Meaning Violet	#66508E
The Light School	 Growth Green	#607A32
Community Service	 Service Amber	#C58A32
Governance	 Stewardship Green	#334D3C
Finance & Fundraising	 Foundation Bronze	#7A5A1E
Internal Communications	 Radiant Magenta	#9B3F76
Branding & Outreach	 Jewel Emerald	#2E8556

SHEET 1 · LOW VISION AND CONTRAST SENSITIVITY

Condition responded to: reduced visual acuity and contrast sensitivity, including age-related change, cataract, and partial sight. What helps is not different hues but stronger separation between figure and ground. WCAG AA requires 4.5:1 for body-size text; seven of the ten base colors already pass on the ivory ground and three receive a deepened twin for text duty. The base color remains the brand color everywhere else.

HOME	BASE VS IVORY	HIGH-CONTRAST HEX	HC VS IVORY	STATUS
Research	6.61 : 1	 #1A6073 (unchanged)	6.61 : 1	passes as is
Technology, AI & Info. Systems	6.38 : 1	 #075BA8 (unchanged)	6.38 : 1	passes as is
Experience Arch. & World Building	3.73 : 1	 #B8502C	4.64 : 1	deepened twin
Lumina Lingua	6.31 : 1	 #66508E (unchanged)	6.31 : 1	passes as is
The Light School	4.53 : 1	 #607A32 (unchanged)	4.53 : 1	passes as is
Community Service	2.78 : 1	 #966926	4.51 : 1	deepened twin
Governance	8.64 : 1	 #334D3C (unchanged)	8.64 : 1	passes as is
Finance & Fundraising	5.92 : 1	 #7A5A1E (unchanged)	5.92 : 1	passes as is
Internal Communications	5.82 : 1	 #9B3F76 (unchanged)	5.82 : 1	passes as is
Branding & Outreach	4.25 : 1	 #2C8053	4.54 : 1	deepened twin

Service Amber at 2.78:1 confirms the orange-on-ivory concern from the color review; its role as a warm ground with heavy supporting colors stands, and its deepened twin #966926 carries any text duty. Luminara Gold #F4BE3B measures 1.60:1 against ivory and never carries text on light grounds; on Midnight it measures 10.12:1 and shines. Document ink #111a4a runs 15.42:1 on ivory.

ON DARK GROUNDS

The same eyes need the same separation on Midnight Blue #030F58 and video grounds, where the base colors run too deep. Each color lifts toward white by eighteen percent; the 3:1 large-text threshold is computed against Midnight. Body text on dark grounds is always Ivory, never a division color; these serve headings, icons, chips, and lower thirds.

HOME	DARK-GROUND HEX	VS MIDNIGHT	LARGE-TEXT AA (3:1)
Research	 #437D8C	3.75 : 1	passes
Technology, AI & Info. Systems	 #3479B8	3.77 : 1	passes
Experience Arch. & World Building	 #D97957	5.62 : 1	passes
Lumina Lingua	 #8270A2	3.95 : 1	passes
The Light School	 #7D9257	5.05 : 1	passes
Community Service	 #CF9F57	7.21 : 1	passes
Governance	 #586D5F	3.11 : 1	passes, barely; prefer icon use over type
Finance & Fundraising	 #927846	4.12 : 1	passes
Internal Communications	 #AD628F	4.06 : 1	passes
Branding & Outreach	 #549B74	5.21 : 1	passes

SHEET 2 • RED-GREEN COLOR VISION DEFICIENCY

Condition responded to: deuteranopia and deuteranomaly (green cone absent or shifted) and protanopia and protanomaly (red cone absent or shifted), together the most common forms: roughly eight percent of men and half a percent of women. Red-green hue distinctions collapse; what survives is luminance, the lightness channel. A pair is safe when its lightness separation is large; it is risky when two hues differ mainly in redness or greenness at similar lightness.

PAIR ON THE RED-GREEN AXIS		PAIR CONTRAST	READING
 Radiant Magenta	 Jewel Emerald	1.37 : 1	the flagged axis: near-twins under deutan vision
 Jewel Emerald	 Growth Green	1.06 : 1	near-identical lightness; confusable for all viewers at small sizes
 Embodied Coral	 Growth Green	1.21 : 1	classic red vs green at similar lightness
 Embodied Coral	 Service Amber	1.34 : 1	adjacent warms; rely on labels
 Foundation Bronze	 Stewardship Green	1.46 : 1	borderline; rarely co-plotted
 Service Amber	 Growth Green	1.63 : 1	acceptable separation

The response, and what it deliberately is not. Deepening Jewel Emerald was tested and makes the magenta pair worse, not better: magenta sits below emerald in lightness, so separation must come from lifting the green, and Drew's magenta is never altered. The response is a **chart accent**: in data graphics, org charts, and any context where Internal Communications and Branding & Outreach (or the two deep greens) appear side by side, Branding & Outreach plots as  #45A873 (Emerald Chart Accent). Pair contrast against magenta rises from 1.37:1 to 2.11:1 and against Growth Green from 1.06:1 to 1.64:1. Brand surfaces keep Jewel Emerald #2E8556 unchanged; the accent exists only where colors must be told apart at a glance. Where even that is not enough, the governing rule on the last sheet applies.

SHEET 3 • BLUE-YELLOW COLOR VISION DEFICIENCY

Condition responded to: tritanopia and tritanomaly (blue cone absent or shifted), the rare form: on the order of one in ten thousand, affecting men and women equally. Blue-yellow distinctions collapse; blues, teals, and violets draw together, as do yellows and warm midtones. The computed pair contrasts on the institute's blue axis:

PAIR ON THE BLUE-YELLOW AXIS		PAIR CONTRAST	READING
 Interface Blue	 Inquiry Teal	1.04 : 1	near-twins under tritan vision
 Meaning Violet	 Interface Blue	1.01 : 1	identical lightness; hue is the only difference
 Meaning Violet	 Inquiry Teal	1.05 : 1	near-twins under tritan vision
 Radiant Magenta	 Meaning Violet	1.08 : 1	draws together as blue perception fades
 Royal Blue	 Stewardship Green	1.13 : 1	deep pair; rarely co-plotted
 Luminara Gold	 Service Amber	1.74 : 1	acceptable separation

The response. The cool cluster (Interface Blue, Inquiry Teal, Meaning Violet, Radiant Magenta) sits within seven percent of one lightness value, so no hue adjustment can separate it for tritan vision without leaving the brand; the separation must come from lightness steps and labels. Where two or more of the cluster are co-plotted, plot them at different lightness levels: take the darker member from this sheet's base value and the lighter member from the dark-ground column on Sheet 1 (a computed lift of eighteen percent). With one exception the cluster rarely shares a chart; the labels rule carries the rest.

SHEET 4 • COMPLETE COLOR BLINDNESS

Condition responded to: achromatopsia, the absence of color perception: roughly one in thirty thousand people, who see the palette as pure luminance. This sheet shows the gray each color actually presents. Where two grays match, those colors are indistinguishable, and only position, label, shape, or pattern separates them.

HOME	COLOR	PRESENTS AS	GRAY
Community Service	 Service Amber	 #969696	lightest
Experience Arch. & World Building	 Embodied Coral	 #7F7F7F	light
Branding & Outreach	 Jewel Emerald	 #767676	light-mid
The Light School	 Growth Green	 #717171	mid
Internal Communications	 Radiant Magenta	 #616161	merged band
Finance & Fundraising	 Foundation Bronze	 #5F5F5F	merged band
Technology, AI & Info. Systems	 Interface Blue	 #5B5B5B	merged band
Lumina Lingua	 Meaning Violet	 #5B5B5B	merged band
Research	 Inquiry Teal	 #585858	merged band
Governance	 Stewardship Green	 #474747	darkest

The response. Five of the ten colors land in one merged gray band (#585858 to #616161): to an achromatopic reader, Research, Technology, Lumina Lingua, Finance, and Internal Communications are the same color. No palette that keeps these brand hues can fix that in gray; the fix is structural and already adopted as the governing rule: every color-coded element carries its label, and printed materials stay legible photocopied in black and white, which is also the test any staff member can run without tools.

SHEET 5 • PHOTSENSITIVITY AND VISUAL STRESS

Condition responded to: light sensitivity and pattern-related visual stress, including migraine and post-concussion sensitivity. What helps is a softened ground, reduced saturation over large areas, and no high-frequency contrast patterns. The document ivory #FBF7EA already sits below pure white; for large fields, the palette softens to these tints, blended three quarters of the way toward ivory. Tints are grounds only; they never carry meaning alone and never carry text without an ink-dark foreground.

HOME	SOFT TINT	INTENDED USE
Research	 #C3D1CC	Section tint
Technology, AI & Info. Systems	 #BED0DA	Section tint
Experience Arch. & World Building	 #F0D0BC	Section tint
Lumina Lingua	 #D6CDD3	Section tint
The Light School	 #D4D8BC	Section tint
Community Service	 #EEDCBC	Section tint
Governance	 #C9CCBE	Row shading
Finance & Fundraising	 #DBD0B7	Row shading
Internal Communications	 #E3C9CD	Row shading
Branding & Outreach	 #C8DAC5	Row shading

Full-saturation Luminara Gold stays an accent at this sheet's sizes, never a full-page ground; animated or flashing brand elements are not used in any material, on any surface.

THE GOVERNING RULE: NEVER COLOR ALONE

Any meaning a color carries is also carried by a word, shape, or pattern. This rule governs every sheet above. Accessibility in this system is achieved first by labeling and iconography, and only second by the luminance engineering on these sheets; no chart, map, badge, or document in the institute's materials may rely on hue as the sole carrier of information. This is the accessibility floor beneath the palette, adopted alongside the every-page identity rule.

Prevalence figures follow published vision-science reporting; pair values and gray equivalents are computed from the hex values by WCAG 2.1 relative luminance. The widely used Okabe-Ito colorblind-safe palette served as the external reference standard when testing the chart accent. Adoption path: these sheets go to Drew Stone for color review; adopted values then join the Branding Style Sheet as application rows. Until then, every value on this page is a computed proposal.